

Uncertainty-driven and Adversarial Calibration Learning for Epicardial Adipose Tissue Segmentation

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Abstract—Epicardial adipose tissue (EAT) is a type of visceral fat that can secrete large amounts of adipokines to affect the myocardium and coronary arteries. However, segmenting EAT is challenging due to the low contrast between EAT and pericardial effusion and the presence of motion artifacts. We propose a novel feature latent space multilevel supervision network (SPDNet) with uncertainty-driven and adversarial calibration learning to enhance segmentation for more accurate EAT volume estimation. The network first addresses the blurring of EAT edges due to the medical images in the open medical environments with low quality or out-of-distribution by modeling the uncertainty as a Gaussian distribution in the feature latent space, which using its Bayesian estimation as a regularization constraint to optimize SwinUNETR. Second, an adversarial training strategy is introduced to calibrate the segmentation feature map and considers the multi-scale feature differences between the uncertainty-guided predictive segmentation and the ground truth segmentation, synthesizing the multi-scale adversarial loss directly improves the ability to discriminate the similarity between organizations. Experiments on the real-world clinical cohort EAT dataset show that the proposed network outperforms mainstream models, validating that uncertainty-driven and adversarial calibration learning can be used to provide additional information for modeling multi-scale ambiguities.

Keywords—component; Epicardial adipose tissue segmentation; Uncertainty-driven; Adversarial calibration; Bayesian estimation

I. INTRODUCTION

Epicardial adipose tissue (EAT) resides within both the atrioventricular and interventricular grooves, enveloping a considerable portion of the cardiac vasculature. EAT functions as a metabolically active endocrine and paracrine organ, releasing significant quantities of both anti-inflammatory and pro-inflammatory adipokines. These bioactive molecules exert influence over the dynamic equilibrium of energy-glycolipid metabolism, as well as cardiac structure and function [1, 2]. EAT is closely associated with the development of coronary heart disease, thus it is a biomarker for a class of risk factors. More findings propose that EAT serves as an independent parameter that is expected to be a surrogate biomarker for cardiovascular risk estimation [3, 4]. Nevertheless, the quantitative imaging assessment of EAT with intricate topology is a challenging task. Since EATs have different physiologic and pathologic properties at different locations, this leads to two major problems for the imaging physician in accurately quantifying their statistical differences: Primarily, the heart is a continually pulsating and contracting muscular organ that plays a pivotal role in pumping blood to sustain proper bodily function. This constant motion produces artifacts, leading to the indistinct delineation of EAT edges from the surrounding tissues. Secondly, there exists a challenge in distinguishing EAT from pericardial effusion, as both may manifest very similar signal intensities in MRI [5]. Consequently, this makes the

difference in their grayscale values small and increases increasing the difficulty of visual differentiation. The above factors lead to inefficient and cognitively subjective manual labeling assessments. Thus, to reduce the burden of quantitative assessment of EAT using image processing techniques for automated operations can provide objective and rapid quantitative assessment, which greatly improves the efficiency of imaging physicians.

In the field of medical image segmentation, most of the current mainstream semantic segmentation methods use Vision Transformer [6] or U-Net [7] architectures, focusing on the learning of deterministic features, e.g. [8-10]. Regrettably, these methodologies overlook the influence of data or model uncertainty issues on segmentation outcomes, encompassing factors like motion artifacts and cognitive ambiguities. So, there may be certain model performance bottlenecks for segmentation problems with such uncertainties. Although there have been some approaches that consider the reliability of segmentation results and introduce uncertainty estimation, where uncertainty is directly or indirectly represented as network parameter distributions or feature latent space distributions in the model, these models are based on learning strategies for a parameter or feature distributions do not consider calibrating the modeled uncertainty [11-13], and lack the capability to further utilize uncertainty to guide the segmentation performance improvement. Thus, we proposed an uncertainty quantification and calibration framework for feature latent space distribution representation for EAT segmentation.

The main contributions of this study are as follows: (1) The Probabilistic Net models data uncertainty by employing a Gaussian distribution within the feature latent space. Feature vectors sampled from this distribution are incorporated into the segmentation feature maps layer by layer and by using quantized uncertainty constraints to suppress segmentation performance degradation due to motion artifacts. (2) The Adversarial training strategy was used to calibrate the segmentation feature maps, and a multi-scale loss optimization model of Discriminator Net was introduced into the objective function to enhance the model's ability to capture the feature differences at each level, to better distinguish similar tissues with low contrast. (3) The model proposed in this study was validated on an open data set for real-world applications, and the efficacy of uncertainty-driven and adversarial calibration methodologies underwent evaluation in self-constructed clinical cohort segmentation tasks. The results show that the proposed method achieves significant improvements in EAT segmentation, which provides strong support for early diagnosis and treatment of coronary artery disease.

II. METHODS

Ours approach is shown in Fig.1, image and image + Ground Truth are first modeled by Priori Net and Posterior Net in Probabilistic Net to construct a priori feature latent space and a posterior feature latent space, respectively. Next, we sample a random feature vector Z from a posterior feature latent space and inject it into the Segmentor Net, which is fused with the Image features extracted by the Segmentor Net to each layer to generate the corresponding segmentation maps and compute the reconstruction loss. Lastly, we employ an adversarial training

strategy to train the model through a Discriminator Net with multiscale loss to compensate for the lack of ability to perform direct learning of multiscale space constraints during the end-to-end training process to calibrate the segmentation feature maps.

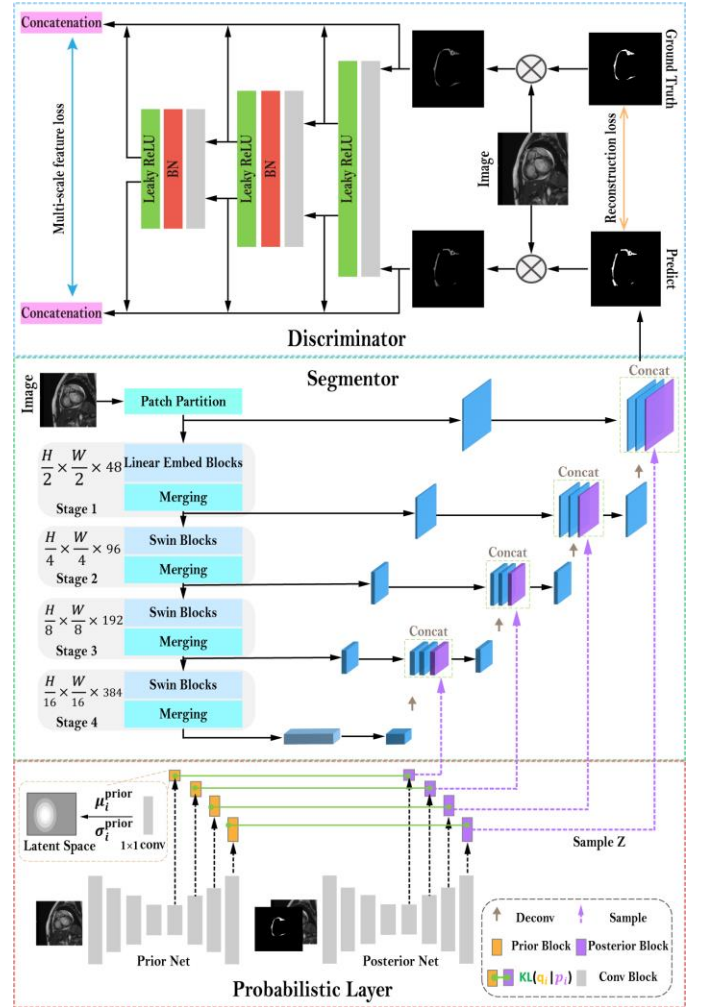


Fig. 1 (1) Segmentor Net is used to extract features from the input image and can fuse random feature vectors sampled from (2) Probabilistic Net, and finally get a coarse segmentation. (3) Discriminator Net fuses the coarse segmentation and the true labeling with the input image, respectively, and then computes the loss from these two different inputs.

A. Segmentor Net

The backbone network SwinUNETR used an encoder-decoder paradigm and has an overall U-shaped structure. Tiny Swin Transformer Blocks (the number of Swin Transformer Blocks for the four stages is [2, 2, 6, 2]) and convolutional layers with the number of channels [48, 96, 192, 384] are used in our approach. This block yields an initial coarse segmentation result, and based on this result, we calculate the reconstruction loss once. The original SwinUNETR loss function comprises the cross-entropy loss function, denoted as $L_{ce}(\hat{y}_i, y_i) = \sum_{i=0}^N y_i \ln \hat{y}_i + (1 - y_i) \ln(1 - \hat{y}_i)$. To capture small lesion features, we introduced the Dice loss, $L_{dice}(\hat{y}_i, y_i) = 1 - \frac{2 \sum_{i=0}^N \hat{y}_i y_i}{\sum_{i=0}^N \hat{y}_i + \sum_{i=0}^N y_i}$. These two losses are

combined using an adjustable parameter to form a new loss function, denoted as L_{rec} :

$$L_{rec}(\hat{y}_i, y_i) = -\alpha L_{ce}(\hat{y}_i, y_i) + (1 - \alpha) L_{dice}(\hat{y}_i, y_i). \quad (1)$$

In the above equation, where $i \in \{0, 1\}$ represents the category (0 for background, 1 for epicardial fat), $\hat{y}_i$ denotes the prediction for category i , and y_i denotes the ground truth of class. The parameter α is artificially set to regulate the ratio between the cross-entropy loss and the Dice loss. In experiments, we find the best performance at $\alpha = 0.6$.

B. Probabilistic Net

The primary function of the a priori net is to model the image X . At the decoding stage, multiple a priori probability distributions $p_i(z_i|x_i)$, are obtained. Each position in these distributions corresponds to a segmentation variable. These are Gaussian distributions with mean μ_i^{prior} and variance $\sigma_i^{\text{prior}} \in \mathbb{R}^{H_i \times W_i}$ (where $i \leq L$, and L is the number of layers of the decoder). We compute the μ_i^{prior} and σ_i^{prior} for the feature latent space at each scale by 1×1 convolution. The a priori net responsible for computing spatial features of a given input image X at different levels. In a priori net, the latent hierarchy at each scale undergoes an additional step: sampling $z_i \in \mathbb{R}^{H_i \times W_i}$ from the low-dimensional latent space constructed via μ_i^{prior} and σ_i^{prior} , which is connected to the features of the input U-Net decoder and then up-sampled in the usual way. By combining the deterministic feature extractor and latent space sampling, we achieve effective utilization of uncertainty at different scales. The latent variables z_i sampled from the corresponding Gaussian distribution in the latent space and the prior distribution P are formulated as follows:

$$z_i \sim \mathcal{N}(\mu_i^{\text{prior}}(x_i), \sigma_i^{\text{prior}}(x_i)) = p(z_i|x_i). \quad (2)$$

$$P(z_0, \dots, z_i|x_0, \dots, x_i) = p(z_0|x_0) \cdot \dots \cdot p(z_i|x_i). \quad (3)$$

The posterior net models the variational a posterior $Q(\cdot|X, Y)$ by requiring inputs of both X and Y . In our case, Y is the segmentation label, and X is the input image. Its role is intended to help maximize the evidence lower bound (ELBO) of the likelihood $p(Y|X)$. In the posterior net, we compute the feature mean $\mu_i^{\text{post}} \in \mathbb{R}^{H_i \times W_i}$ and variance $\sigma_i^{\text{post}} \in \mathbb{R}^{H_i \times W_i}$ at each scale and model a Gaussian distribution with these means and variances. During the training period, the sampled samples z_i are fed into the Segmentor Net. The purpose of this step is to reduce the interference of motion artifacts on the network by using uncertainty constraints to guide the loss function L_{rec} during the training phase. In the posterior net, the posterior latent variable z_i and the posterior distribution P are formulated as follows:

$$z_i \sim \mathcal{N}(\mu_i^{\text{post}}(x_i, y_i), \sigma_i^{\text{post}}(x_i, y_i)) = q(z_i|x_i, y_i), \quad (4)$$

$$Q(z_0, \dots, z_i|(x_0, \dots, x_i), (y_0, \dots, y_i)) = \\ (z_0|x_0, y_0) \cdot \dots \cdot q(z_i|x_i, y_i). \quad (5)$$

In addition, there is a Kullback-Leibler divergence

$D_{KL}(Q||P) = E_{z \sim Q}[\log Q - \log P]$, which brings P and Q closer together. We also use a weight parameter β to participate in the computation of the reconstruction loss during the coarse segmentation loss computation (We found that the proposed method works robustly if $\beta = 10$), and our loss function formula is shown below:

$$L_{ELBO} =$$

$$E_{z \sim Q}[L_{rec}(S(X, z), Y)] + \beta \cdot \sum_{i=0}^L E_{z_i \sim Q} D_{KL}(q_i(z_i|x_i, y_i) || p_i(z_i|x_i)). \quad (6)$$

C. Discriminator Net

Discriminator Net has two different inputs: the coarse segmented image $S(X, z)$ fused with the input image X and the segmented label Y fused with the input image X , denoted as $D(X * S(X, z))$ and $D(X * Y)$. The Discriminator Net employs a multilayer structure to extract hierarchical features and compute a multiscale loss L_1 . This loss makes full use of the different scale of features, allowing the model i.e. to learn global and local features more efficiently and to calibrate the segmented feature map further. Training of the Discriminator Net occurs in an alternating manner with the Segmentor Net through backpropagation of the multiscale loss L_1 . Through continual training and parameter updates, the Discriminator Net and Segmentor Net will become more and more powerful and gradually reach a balance. In Discriminator Net we use Mean Absolute Error (MAE) L_{mae} for multi-scale feature loss, so our Discriminator Net loss is as follows:

$$L_{mae} = \frac{1}{C} \sum_{i=1}^C \left\| (d_i(X * Y), d_i(X * S(X, z))) \right\|_1. \quad (7)$$

In the above equation, C is the total number of layers of the Discriminator Net, and d_i represents the features extracted from the i th convolutional layer.

In the following expression, θ_S and θ_D represent the parameters of the generator and discriminator, respectively, and K is the batch size of the input image. To better the optimization of our objective function, we integrate the ELBO loss with the discriminator training loss. Thus, the total loss of our model is expressed as follows:

$$\min_{\theta_S} \max_{\theta_D} L_{total}(\theta_S, \theta_D) \\ = \frac{1}{K} \sum_{i=1}^K L_{mse}(D(X * Y), D(X * S(X, z))) + L_{ELBO}. \quad (8)$$

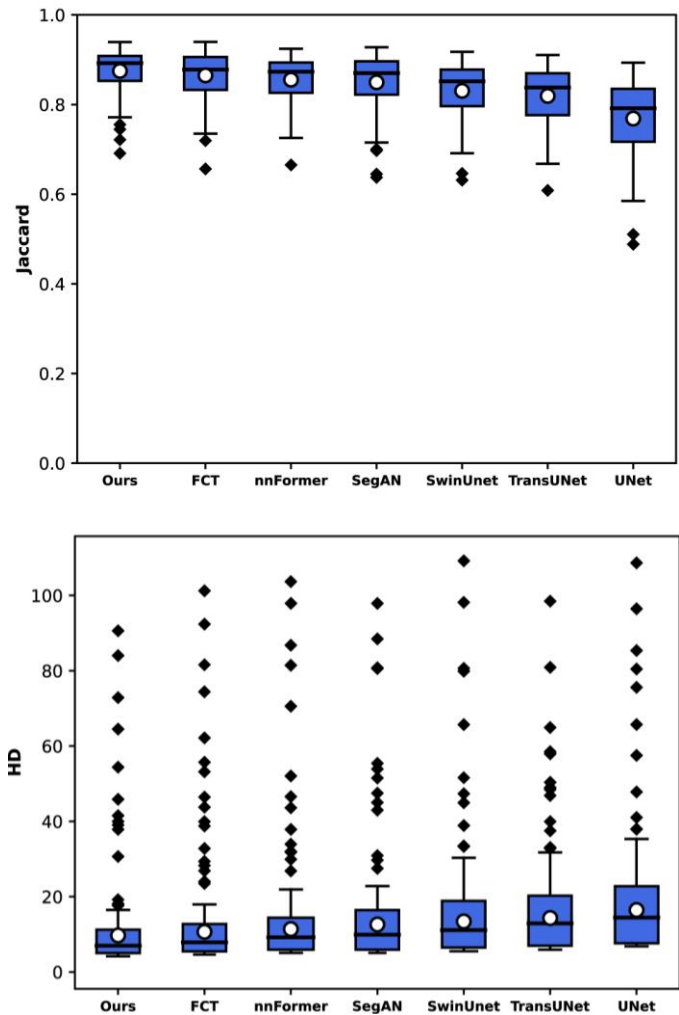
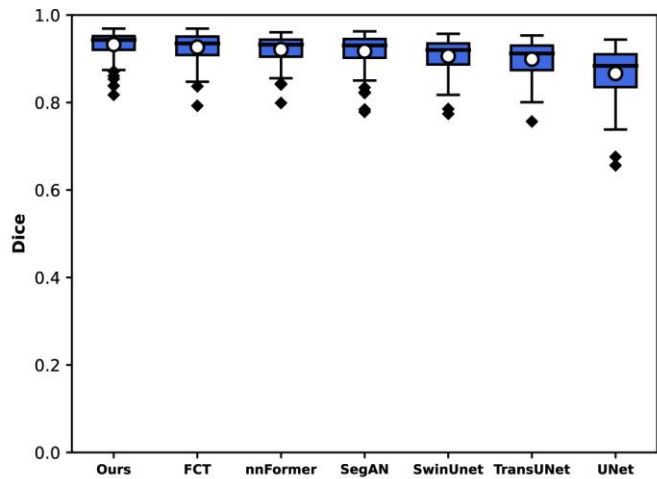
III. EXPERIMENTS

In this study, we conducted a comprehensive performance evaluation of the model using the real-world epicardial fat dataset. We deliberately selected six contemporary and leading medical image segmentation methods: SegAN [14], FCT [15], nnFormer [16], SwinUnet [17], TransUNet [18], and U-Net [7] for comparative experiments. To assess the segmentation performance, three metrics were employed: Dice score, Jaccard

(Jaccard similarity coefficient), and Harsdorff distance (HD). The model underwent training for 100 epochs using Adam's optimization function, with a batch size set to 8 and an initial learning rate of 0.0001. Throughout the training process, we utilized an NVIDIA 3090 graphics card and enhanced the model's performance with data augmentation techniques, including random rotations, flipping samples on either the x- or y-axis, and introducing random angular skews.

A. Comparative experiments

According to Table1, we conducted a comprehensive comparison of our model with six other models. The results demonstrate that our model achieves optimal scores on all three evaluation metrics. Specifically, the Dice score reaches 0.933, the Jaccard coefficient is 0.875, and the Harsdorff distance is 9.696mm. Additionally, we visually represented the distribution of the evaluation metric scores through box-and-line plots in Fig. 2. The results clearly indicate that, compared to the other methods, the model significantly reduces the number of outliers predicted for the three metrics. These results fully showcase the excellent performance of our model in medical image segmentation tasks. Observing Fig.3, we note a significant segmentation deficiency in other methods, particularly in cases of motion blur. Additionally, in images where epicardial fat and pericardial effusion coexist, other methods encounter difficulty in differentiation. This underscores the model's effective mitigation of the effects of motion blur and its enhanced ability to accurately distinguish epicardial fat from pericardial effusion.



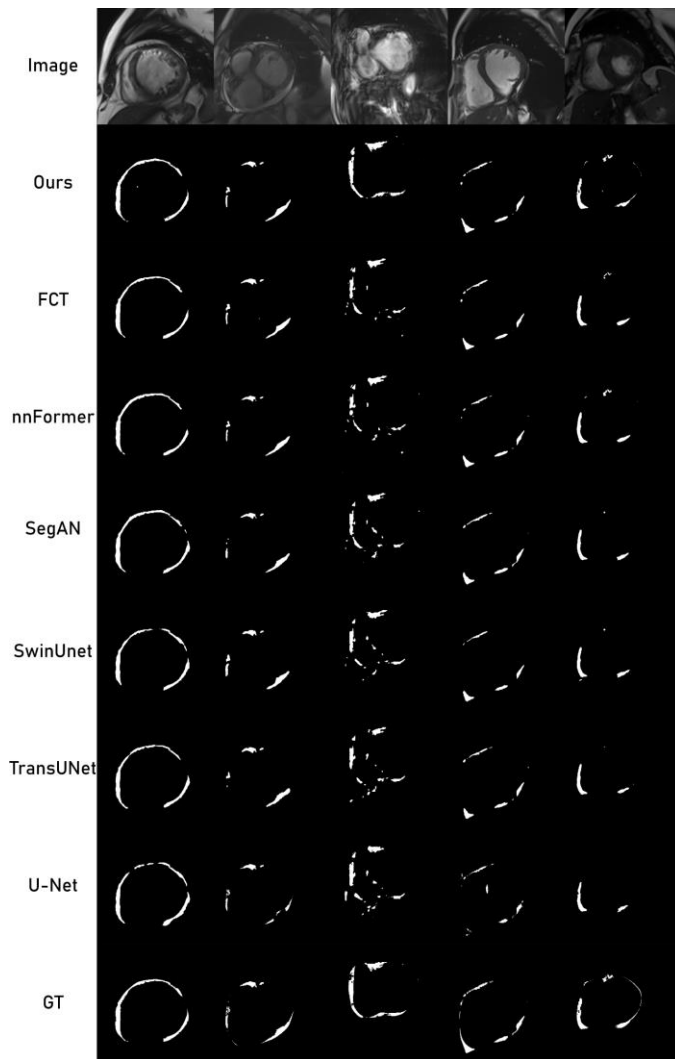


Fig. 3 The visualized of the epicardial fat segmentation results, with each row in the first column being a different MRI slice of the patient's heart.

Table 1 Comparative experiment results with the segmentation methods in the table were evaluated on the epicardial fat dataset with black bolding representing the best performance.

Methods	Dice score	Jaccard	HD (mm)
U-Net[7]	0.866	0.771	13.721
TransUNet[28]	0.899	0.822	12.617
SwinUnet[27]	0.906	0.831	12.503
SegAN[21]	0.917	0.849	11.368
nnFormer[26]	0.921	0.855	10.683
FCT[25]	0.927	0.865	10.356
Ours	0.933	0.875	9.696

B. Ablation experiments

The performance improvement of our algorithm lies in the introduction of two key nets: the Probabilistic Net and the Discriminator Net. To assess the effectiveness of these nets, we conducted exhaustive ablation experiments, except for adding or subtracting net, we did not modify other aspects (in Fig. 4). The results of the experiments, as presented in Table 2, indicate a significant enhancement in the model's performance with the expansion of each net. Introducing the Probabilistic Net results in a 0.04 improvement in the Dice coefficient score, a 0.06 improvement in the Jaccard index, and a 0.392 mm improvement in the Harsdorff distance index. This experiment substantiates the efficacy of our enhancements. Similarly, the addition of the Discriminator Net further enhances the performance across each metric.

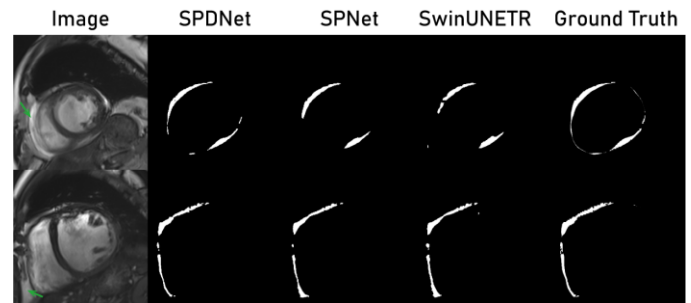


Fig. 4 The first row of case sections illustrates instances with motion artifacts, while the second row showcases sections where pericardial effusion is present in proximity to epicardial adipose tissue. Relevant areas have been highlighted with green arrows for clarity.

Table 2 Performance of ablation experiments with different modules.

Probabilistic Net	Discriminator Net	Dice score	Jaccard	HD (mm)
		0.921	0.857	10.679
√		0.925	0.863	10.287
√	√	0.933	0.875	9.696

IV. CONCLUSIONS

In this study, epicardial fat segmentation faced two major challenges: the difficulty in distinguishing the presence of motion artifacts and EAT from pericardial effusion. In addition, manual labeling of epicardial fat is a time-consuming and laborious task. Therefore, our study aims to propose an efficient automated segmentation model to assist clinicians in risk assessment and thus improve their productivity. Our approach optimizes the Segmentor Net by transforming the blurring of epicardial adipose tissue (EAT) edges induced by motion artifacts into a regularization constraint via the Probabilistic Net. This optimization enhances segmentation performance through the utilization of uncertainty representation. Subsequently, an adversarial training strategy is employed to learn the predictive segmentation maps and ground truth multiscale feature differences by constructing a Discriminator Net with multiscale loss to calibrate the predictive segmentation maps and better

differentiate pericardial effusion from epicardial fat. The proposed model was experimentally validated on the real-world epicardial fat dataset. The experimental results show that the proposed model has achieved significant performance improvement, which provides a reliable auxiliary tool for clinicians to accurately segment epicardial fat.

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